

SAP SECURITY ADMINISTRATION, OPERATIONS & TROUBLESHOOTING

Chapter 2 – PFCG, SU24 & Advanced Role Administration

Q201. What is PFCG?

Interviewer's Objective

The interviewer wants to understand whether you know PFCG as a role administration framework, not just a transaction.

Executive Answer

PFCG (Profile Generator) is SAP's central tool for designing, creating, maintaining, generating, and transporting authorization roles.

It translates business requirements into technical authorizations by combining:

- **Menu structure**
- **Transactions**
- **Authorization Objects**
- **Organizational Levels**
- **Authorization Profiles**

PFCG automatically generates authorization profiles based on the role definition and SU24 authorization proposals.

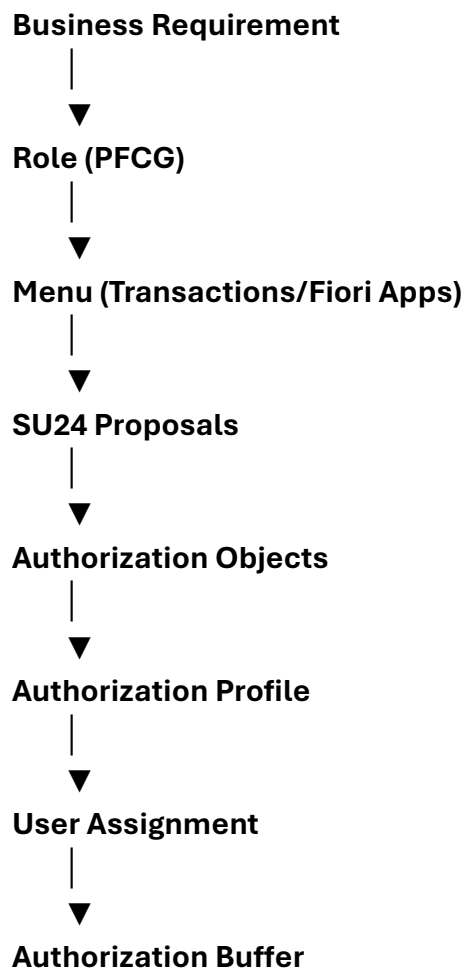
Responsibilities of PFCG

- **Create Single Roles**
- **Create Composite Roles**
- **Create Derived Roles**
- **Generate Authorization Profiles**
- **Assign Users**
- **Maintain Organizational Levels**
- **Transport Roles**
- **Perform User Comparison**

Q202. Explain the internal architecture of PFCG.

Executive Answer

Internally, PFCG follows this logical flow:



PFCG itself does not decide which authorization objects are required. It reads authorization proposals from SU24 and generates the authorization profile accordingly.

Q203. What are the different types of SAP roles?

Executive Answer

SAP supports several role types.

Role Type	Purpose
Single Role	One business function
Composite Role	Collection of Single Roles
Derived Role	Inherits authorizations from a Master Role with different organizational values

Single Role

Contains:

- **Transactions**
- **Authorization Objects**
- **Organizational Levels**

Example

Z_MM_BUYER

Composite Role

Contains multiple Single Roles.

Example

FINANCE_MANAGER

Contains:

- **AP Role**
- **AR Role**
- **Reporting Role**

Composite Roles simplify user assignment.

Derived Role

Inherits:

- **Menu**

- **Authorization Objects**

Changes only:

- **Company Code**
- **Plant**
- **Sales Organization**

Derived Roles reduce maintenance in multi-company environments.

Q204. Explain Master and Derived Roles.

Executive Answer

A Master Role contains:

- **Transactions**
- **Menu**
- **Authorization Objects**

It does not contain organization-specific values.

Derived Roles inherit everything from the Master Role but define their own organizational values.

Example:

Master Role

Z_MM_BUYER_MASTER

↓

Derived Roles

Z_MM_BUYER_US

Z_MM_BUYER_INDIA

Z_MM_BUYER_UAE

Advantages

- **Centralized maintenance**
 - **Consistent authorizations**
 - **Reduced role duplication**
 - **Easier upgrades**
 - **Lower testing effort**
-

Q205. Explain Composite Roles.

Executive Answer

A Composite Role is a logical grouping of multiple Single Roles.

Example

Finance Manager

↓

Contains

- **AP Role**
- **AR Role**
- **Reporting Role**
- **Display Role**

The Composite Role itself does not contain authorization objects; it references the included Single Roles.

Q206. Explain Authorization Profile Generation.

Executive Answer

When a role is generated:

Transactions

↓

SU24



Authorization Objects



Authorization Values



Authorization Profile



User

The generated authorization profile is what SAP evaluates during authorization checks.

Whenever authorization data changes, the role should be regenerated to ensure the profile reflects the latest changes.

Q207. What is SU24?

One of the most frequently asked SAP Security interview questions.

Executive Answer

SU24 maintains authorization proposals for transactions.

It defines:

- **Authorization Objects**
- **Check Indicators**
- **Default Field Values**

When a transaction is added to a role in PFCG, PFCG reads SU24 to determine which authorization objects should be proposed.

Benefits

- Standardized role creation
 - Reduced manual effort
 - Consistent authorizations
 - Easier maintenance
-

Q208. Explain Check Indicators in SU24.

Executive Answer

SU24 provides three main options for authorization checks:

Indicator	Meaning
-----------	---------

Check	Perform authorization check
-------	-----------------------------

Check/Maintain	Check authorization and propose object in PFCG
----------------	--

Do Not Check	Authorization check is disabled for that object
--------------	---

Best Practice

Changes to SU24 should be carefully governed because they affect future role generation.

Q209. How does SU24 integrate with PFCG?

Executive Answer

When a transaction is inserted into a role:

1. PFCG reads SU24.
2. SU24 supplies the authorization proposals.
3. PFCG creates authorization objects.
4. Security administrator maintains required field values.

5. Role is generated.

Without accurate SU24 proposals, role creation becomes inconsistent and more manual.

Q210. Explain Organizational Levels.

Executive Answer

Organizational Levels restrict access based on organizational structures.

Examples:

- **Company Code**
- **Plant**
- **Sales Organization**
- **Purchasing Organization**
- **Controlling Area**

Instead of creating different authorization objects for each location, SAP allows organizational values to vary while reusing the same role design.

Example

Master Role

↓

Purchasing Role

↓

Derived Role

↓

Plant = 1000

Q211. What happens when you regenerate a role?

Executive Answer

Regenerating a role performs the following:

- 1. Reads updated authorization data.**
- 2. Creates or updates the authorization profile.**
- 3. Synchronizes authorization objects.**
- 4. Prepares the role for user comparison.**

Regeneration is required whenever authorization data changes.

Q212. What is User Comparison in PFCG?

Executive Answer

User Comparison synchronizes generated authorization profiles with assigned users.

Without User Comparison:

- The role may be updated.**
 - The profile may be generated.**
 - But users may continue using old authorizations until synchronization occurs.**
-

Q213. Important PFCG Tables

Senior interviewers frequently ask this.

Table	Purpose
--------------	----------------

AGR_DEFINE	Role Definition
-------------------	------------------------

AGR_USERS	User Assignments
------------------	-------------------------

AGR_1251	Authorization Objects
-----------------	------------------------------

Table	Purpose
AGR_1252	Organizational Values
AGR_AGRS	Composite Role Assignments
AGR_TEXTS	Role Descriptions
UST12	Authorization Profile Contents
UST04	User Profile Assignments

Q214. Real Production Scenario

Interviewer

"A new transaction was added to a role in PFCG. The role was transported to Production, but users still cannot execute the transaction. What would you check?"

Executive Answer

I would follow a structured troubleshooting process:

1. Verify the transaction was correctly added to the role.
2. Confirm the role was regenerated after the change.
3. Ensure the transport containing the updated role was successfully imported.
4. Perform or verify User Comparison so the updated authorization profile is assigned to users.
5. Ask the user to reproduce the issue and review SU53 or execute STAUTHTRACE.
6. Validate that SU24 proposals included the required authorization objects.
7. Confirm organizational values such as Company Code or Plant are correct.
8. Check whether the transaction also requires additional backend authorizations, RFC permissions, or Fiori-related components if applicable.

Q215. Real Manufacturing Scenario

Interviewer

"A purchasing role is copied to a new manufacturing plant, but users in the new plant still cannot create purchase orders. How would you investigate?"

Executive Answer

My approach would be:

1. Compare the derived role with the master role to ensure inheritance is correct.
 2. Verify that the Plant organizational level has been updated appropriately in the derived role.
 3. Confirm the role has been regenerated after the organizational value change.
 4. Execute User Comparison for the affected users.
 5. Review SU53 or STAUTHTRACE after reproducing the issue.
 6. Validate Purchasing Organization, Purchasing Group, and Company Code values, as purchase order creation often depends on multiple organizational levels.
 7. Confirm with the MM functional consultant that configuration supports the new plant and purchasing organization.
-

Architect's Insight

One of the strongest interview answers you can give is:

"PFCG does not create authorizations by itself. It consumes authorization proposals from SU24, generates authorization profiles, and assigns those profiles to users. The quality of role design therefore depends heavily on accurate SU24 maintenance."

That demonstrates an understanding of how SAP authorization management works internally rather than simply knowing which transaction to use.

Chapter Summary

By the end of this chapter, you should be comfortable explaining:

- **PFCG architecture**
 - **Single, Composite, and Derived Roles**
 - **Master/Derived role strategy**
 - **Authorization profile generation**
 - **SU24**
 - **Check Indicators**
 - **Organizational Levels**
 - **User Comparison**
 - **AGR_* tables**
 - **Production troubleshooting**
 - **Manufacturing-specific role scenarios**
-

Next Chapter – RFC Security, Background Jobs, Interface Users & Technical Security

This chapter will cover approximately 25–30 advanced interview questions, including:

- **RFC Security (SM59)**
- **Trusted RFC**
- **SNC (Secure Network Communications)**
- **Interface users**
- **Background job security**
- **Batch users**
- **Trusted systems**
- **Gateway security**
- **SAP Router**
- **Technical user governance**
- **Security audit considerations**
- **Production troubleshooting**
- **Architect-level scenarios**

This is another favorite topic for senior interviewers because it combines SAP Security with Basis, integration, and enterprise architecture.